

Oral, Dental, and Craniofacial Health

Teeth, periodontium, saliva, mucosa, microbiome, jaw, swallowing, airway, and measurement engineering

Volume 03L of the Health Atlas Research Suite

Purpose

Treat oral health as integral to nutrition, pain, infection control, communication, appearance and systemic health while connecting dental devices and assays to explicit validation requirements.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This volume does not provide dental treatment plans, antibiotic regimens, airway diagnosis from dental imaging or emergency management of deep facial infection. Imaging and salivary/AI tests require a defined clinical question.

How to use this volume

Oral health is the capacity to eat, speak, swallow, breathe, communicate and participate without preventable pain, infection or structural failure. It is not separable from diet, diabetes, immunity, medications, development, airway, social conditions or access. A complete profile combines disease activity, tissue support, saliva, function, symptoms, behavior, systemic context and longitudinal care.

Questions this volume is designed to answer

1. How do mineral balance, biofilm ecology, immunity, saliva and behavior interact?
2. What should a complete oral-health profile measure beyond tooth count?
3. Which dental devices and digital measurements are valid?
4. How do oral disease and systemic disease share causes without overstating causation?
5. How should oral health be integrated into primary care and public health?

Scope

- Tooth, periodontal, salivary, mucosal, microbial, jaw and craniofacial physiology.
- Caries, periodontal, pulpal, infectious, cancer, pain, developmental and tooth-loss states.
- Clinical, radiographic, optical, mechanical, biochemical and functional measurement.
- Prevention, restorative/endodontic/periodontal, orthodontic, surgical and assistive care.
- Medical-dental integration, stewardship and population systems.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

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Executive synthesis

Oral health is the capacity to eat, speak, swallow, breathe, communicate and participate without preventable pain, infection or structural failure. It is not separable from diet, diabetes, immunity, medications, development, airway, social conditions or access. A complete profile combines disease activity, tissue support, saliva, function, symptoms, behavior, systemic context and longitudinal care.

Central organizing principle

Treat oral health as integral to nutrition, pain, infection control, communication, appearance and systemic health while connecting dental devices and assays to explicit validation requirements.

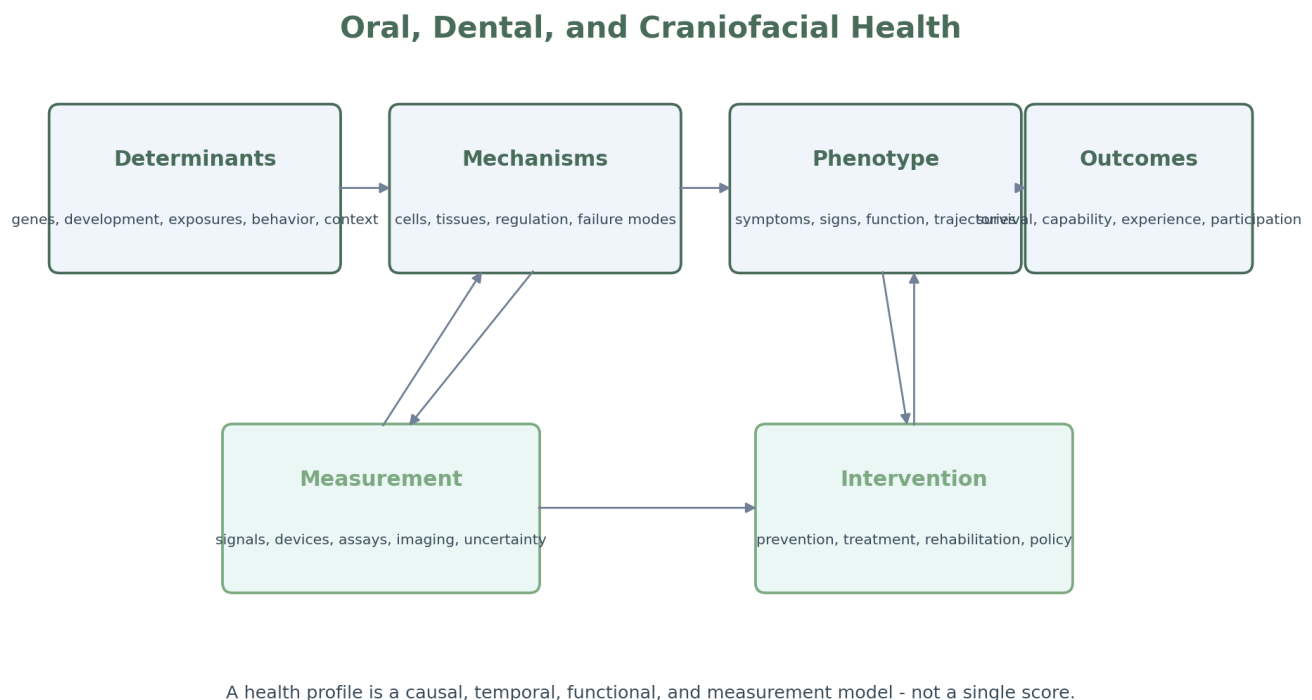


Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.

- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Rapid facial/neck swelling, breathing or swallowing difficulty, uncontrolled bleeding, spreading infection, dental trauma with airway/neurologic injury or suspicious persistent oral lesion requires prompt professional evaluation. Dental devices and imaging involve radiation, biocompatibility, infection-control and airway risks.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Early traditions	Early dentistry combined extraction, restoration, hygiene and dietary observations.	Created observational language for oral and craniofacial health and practical care.	Mechanisms and categories mixed empirical findings with culture-specific theory.
Anatomical and pathological era	Anatomy, microscopy and pathology connected teeth, pulp, periodontium and craniofacial structures.	Localized structure and tissue change.	Static anatomy poorly represented dynamics, perception, function and context.
Instrumental and laboratory era	Radiography, local anesthesia, fluoride, antibiotics and standardized restorative/surgical methods transformed care.	Turned previously subjective phenomena into reproducible signals and images.	Thresholds and reference populations sometimes became mistaken for health itself.
Molecular and therapeutic era	Biofilm ecology, immunology, materials science, implants and molecular oncology expanded mechanisms and treatment.	Connected cells, genes, microbes and signaling to targeted prevention and treatment.	Mechanistic specificity often exceeded evidence of patient-important benefit.
Current multimodal era	Digital scanning, CBCT, optical diagnostics, salivary omics and integrated prevention create new measurement pathways.	Combines imaging, physiology, function, patient report, longitudinal sensing and computation.	Bias, interoperability, overdiagnosis, access and real-world utility remain limiting.
Systems interpretation	Oral and craniofacial health is now best modeled across tissue state, neural perception, behavior, environment, participation and care.	Supports primary-home and cross-volume reasoning.	Training, billing and datasets still separate closely coupled phenomena.

2.0.1 Early dentistry combined extraction, restoration, hygiene and dietary observations.

Scientific consequence. Created observational language for oral and craniofacial health and practical care.

What remained unresolved. Mechanisms and categories mixed empirical findings with culture-specific theory.

Evidence anchors: [3,4]

2.0.2 Anatomy, microscopy and pathology connected teeth, pulp, periodontium and craniofacial structures.

Scientific consequence. Localized structure and tissue change.

What remained unresolved. Static anatomy poorly represented dynamics, perception, function and context.

Evidence anchors: [3,4]

2.0.3 Radiography, local anesthesia, fluoride, antibiotics and standardized restorative/surgical methods transformed care.

Scientific consequence. Turned previously subjective phenomena into reproducible signals and images.

What remained unresolved. Thresholds and reference populations sometimes became mistaken for health itself.

Evidence anchors: [3,4]

2.0.4 Biofilm ecology, immunology, materials science, implants and molecular oncology expanded mechanisms and treatment.

Scientific consequence. Connected cells, genes, microbes and signaling to targeted prevention and treatment.

What remained unresolved. Mechanistic specificity often exceeded evidence of patient-important benefit.

Evidence anchors: [3,4]

2.0.5 Digital scanning, CBCT, optical diagnostics, salivary omics and integrated prevention create new measurement pathways.

Scientific consequence. Combines imaging, physiology, function, patient report, longitudinal sensing and computation.

What remained unresolved. Bias, interoperability, overdiagnosis, access and real-world utility remain limiting.

Evidence anchors: [3,4]

2.0.6 Oral and craniofacial health is now best modeled across tissue state, neural perception, behavior, environment, participation and care.

Scientific consequence. Supports primary-home and cross-volume reasoning.

What remained unresolved. Training, billing and datasets still separate closely coupled phenomena.

Evidence anchors: [1]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Tooth mineral and pulp biology

Core principle. Enamel, dentin, cementum and living pulp form a mineralized organ exposed to mechanical and microbial challenge.

Control architecture. Mineral saturation, saliva, fluoride, diet frequency, dentin-pulp defense and development.

Failure modes. Demineralization/caries, fracture, sensitivity, pulpitis or developmental defect.

Observable consequences. Clinical/radiographic lesion, vitality, symptoms and structural integrity.

Cross-system coupling. Nutrition, microbiome, pain and life-course development.

Evidence anchors: [3,4,5]

3.0.2 Periodontium and alveolar bone

Core principle. Gingiva, ligament, cementum and bone attach teeth and respond to biofilm and load.

Control architecture. Barrier/immune control, collagen turnover, occlusal forces, smoking/metabolism and hygiene.

Failure modes. Gingivitis, periodontitis, attachment/bone loss, mobility and tooth loss.

Observable consequences. Probing, bleeding, radiographs, mobility and risk/trajectory.

Cross-system coupling. Immune, diabetes, smoking, cardiovascular and pregnancy systems.

Evidence anchors: [3,4,13]

3.0.3 Saliva and oral chemical environment

Core principle. Saliva lubricates, buffers, remineralizes, digests, defends and supports taste/speech/swallowing.

Control architecture. Glands, autonomic control, hydration, medications, circadian state and systemic disease.

Failure modes. Xerostomia, caries, infection, dysphagia, taste change and mucosal injury.

Observable consequences. Flow, pH/buffer, symptoms, gland imaging and oral outcomes.

Cross-system coupling. Renal/fluid, autonomic, pharmacology, sensory and nutrition systems.

Evidence anchors: ^[3,6]

3.0.4 Oral microbiome and ecological caries/periodontal model

Core principle. Microbial communities exist normally; disease emerges when ecology shifts under diet, inflammation, hygiene and host factors.

Control architecture. Substrate frequency, pH, biofilm structure, immunity, saliva and niches.

Failure modes. Caries, periodontal disease, halitosis, infection and possible systemic dissemination.

Observable consequences. Clinical disease, plaque ecology and microbiology only when it changes management.

Cross-system coupling. GI, immune, infection and public-health systems.

Evidence anchors: ^[3,7,8]

3.0.5 Mucosa and barrier immunity

Core principle. Oral epithelium tolerates food/microbes while resisting trauma, pathogens and carcinogens.

Control architecture. Turnover, saliva, local immunity, microbiome and exposure repair.

Failure modes. Ulceration, candidiasis, immune lesions, premalignancy or cancer.

Observable consequences. Inspection/palpation, biopsy, microbial tests and systemic evaluation.

Cross-system coupling. Skin/immune, infection, cancer and nutrition.

Evidence anchors: ^[3,10,11]

3.0.6 Mastication, occlusion and temporomandibular function

Core principle. Teeth, jaw joints, muscles and sensorimotor control prepare food and support speech.

Control architecture. Motor control, periodontal/proprioceptive feedback, joint mechanics, load and learned behavior.

Failure modes. Pain, clicking/locking, bruxism, wear, fracture, inefficient chewing or headache.

Observable consequences. Function, range, muscle/joint exam, dental wear and imaging selectively.

Cross-system coupling. Musculoskeletal, nervous, sleep and psychosocial systems.

Evidence anchors: [3,14,15]

3.0.7 Craniofacial growth and airway

Core principle. Neural crest-derived structures grow under genetic, hormonal, functional and environmental influences.

Control architecture. Sutures, cartilage/bone growth, eruption, breathing/swallowing and muscle forces.

Failure modes. Cleft, malocclusion, airway compromise, developmental asymmetry or feeding/speech problems.

Observable consequences. Growth/occlusion, imaging, airway, hearing, speech and feeding.

Cross-system coupling. Maternal/fetal, respiratory, sensory and development systems.

Evidence anchors: [20,21,22]

3.0.8 Swallowing and oral motor control

Core principle. Coordinated sensory-motor phases move food safely from mouth to pharynx/esophagus.

Control architecture. Cranial nerves, brainstem/cortex, tongue, saliva, dentition and airway protection.

Failure modes. Dysphagia, aspiration, malnutrition, prolonged meals or communication problems.

Observable consequences. Clinical swallow/oral motor assessment and instrumental study when indicated.

Cross-system coupling. Neurologic, respiratory, GI, nutrition and rehabilitation systems.

Evidence anchors: [16,23]

3.0.9 Oral-systemic and social function

Core principle. Oral health affects nutrition, pain, sleep, speech, appearance, social participation and healthcare access.

Control architecture. Inflammation, infection, function, stigma, cost and shared determinants.

Failure modes. Avoidable disease, delayed care, school/work loss and inequity.

Observable consequences. Clinical status plus pain, diet, communication, participation and access.

Cross-system coupling. Whole-person, public health, social and delivery-system domains.

Evidence anchors: [3,24,36]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Henderson-Hasselbalch buffer concept	$pH = pK_a + \log_{10}([A^-]/[HA])$	Illustrates how salivary buffer ratios influence pH after acid challenge.	Oral biofilm has spatial gradients, multiple buffers and dynamic flow.
Pressure/stress at occlusal contact	$\sigma = F/A$	Concentrated contact increases local stress on teeth and restorations.	Real contacts are dynamic, viscoelastic and geometry-dependent.
Diagnostic odds update	$Posterior\ odds = Prior\ odds \times LR$	Adjunct optical or salivary tests modify, rather than replace, clinical probability.	Likelihood ratios must match population, protocol and disease definition.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within oral, dental, and craniofacial health.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within oral, dental, and craniofacial health.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

Evidence anchors: ^[20]

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Evidence anchors: ^[19]

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

Evidence anchors: ^[15]

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

Evidence anchors: ^[18]

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

Evidence anchors: ^[24]

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

Evidence anchors: ^[24]

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[35]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the oral, dental, salivary, craniofacial or functional disorder, but cross-system processes and functional effects are retained.

5.0.1 Dental caries

Definition and phenotype. Biofilm-mediated, diet-modulated demineralization and remineralization imbalance causing progressive tooth tissue loss.

Pathophysiology. Frequent fermentable carbohydrate lowers plaque pH; saliva/fluoride and behavior modify net mineral balance.

Determinants and modifiers. Diet frequency, fluoride, saliva, hygiene, development, disability, poverty and dental access.

Assessment. Visual-tactile and radiographic assessment with lesion activity and individual risk.

Intervention logic. Fluoride/sealants, dietary/hygiene support, minimally invasive restoration and pulp treatment according to stage.

Known limits and open questions. Risk scores and microbiome tests add limited value without changing prevention; overtreatment of arrested lesions is possible.

Suite links. 11 prevention; 06 social determinants; 14 interventions.

Evidence anchors: [3,4,24]

5.0.2 Periodontal disease

Definition and phenotype. Inflammatory destruction of tooth-supporting tissues driven by dysbiotic biofilm and susceptible host response.

Pathophysiology. Biofilm triggers immune-mediated connective tissue and bone loss; smoking and diabetes alter response/repair.

Determinants and modifiers. Hygiene/access, smoking, diabetes, age, genetics, stress and medications.

Assessment. Probing depth/attachment, bleeding, radiographs, mobility and longitudinal staging/risk.

Intervention logic. Biofilm control, professional debridement, smoking/diabetes management, surgery selectively and maintenance.

Known limits and open questions. Causal links to systemic outcomes are often overstated; shared risk and inflammation must be separated from treatment effects.

Suite links. 03D diabetes; 03A cardiovascular; 03H immune.

Evidence anchors: [3,4,13]

5.0.3 Pulpal and periapical disease

Definition and phenotype. Inflammation or necrosis of dental pulp with possible infection beyond root apex.

Pathophysiology. Caries, cracks or trauma breach defenses; confined pulp pressure and microbial invasion cause pain and bone response.

Determinants and modifiers. Untreated caries, trauma, anatomy, delays and restoration history.

Assessment. Pain pattern, thermal/electric vitality, percussion, radiographs/CBCT selectively and tissue findings.

Intervention logic. Caries removal/restoration, root-canal treatment, drainage/extraction and antibiotics only for systemic/spreading infection.

Known limits and open questions. Vitality tests measure neural response rather than true blood flow; imaging may lag.

Suite links. 08 infection; 14 antimicrobial stewardship.

Evidence anchors: [4,8]

5.0.4 Odontogenic infection and facial-space spread

Definition and phenotype. Dental infection extending into bone or fascial spaces with airway/sepsis risk.

Pathophysiology. Polymicrobial infection follows necrosis or periodontal source and spreads along anatomy.

Determinants and modifiers. Delayed access, immune/metabolic disease, poor source control and AMR.

Assessment. Airway/systemic severity, source localization, imaging for deep spread and cultures in severe/atypical disease.

Intervention logic. Urgent drainage/source control, airway support and targeted antimicrobials.

Known limits and open questions. Pain and swelling can underestimate deep progression.

Suite links. 03B airway; 08 infection; 12 emergency systems.

Evidence anchors: [3,8,35]

5.0.5 Oral mucosal ulcerative and immune disease

Definition and phenotype. Traumatic, infectious, inflammatory, autoimmune, nutritional, drug-related or neoplastic lesions causing pain and barrier loss.

Pathophysiology. Epithelial injury and immune pathways vary by cause.

Determinants and modifiers. Trauma, infection, immune disease, medications, deficiencies, tobacco and systemic illness.

Assessment. Duration/pattern, complete oral/system exam, targeted labs/microbiology and biopsy of persistent/suspicious lesions.

Intervention logic. Remove cause; topical/systemic therapy by mechanism; pain/nutrition support.

Known limits and open questions. Many lesions look alike and empiric steroid/antimicrobial use can obscure diagnosis.

Suite links. 03H immune; 03G hematology; 14 pharmacology.

Evidence anchors: [3,5]

5.0.6 Oral and oropharyngeal cancer

Definition and phenotype. Malignant epithelial and other tumors affecting mouth/throat, function and survival.

Pathophysiology. Mutagenic exposure, HPV for selected sites, clonal evolution, invasion and nodal spread.

Determinants and modifiers. Tobacco, alcohol, areca, HPV, UV for lip, age and care access.

Assessment. Inspection/palpation, biopsy, imaging and stage; dental/nutrition/speech baseline.

Intervention logic. Exposure prevention, surgery/radiation/systemic therapy and rehabilitation/palliative support.

Known limits and open questions. Population visual screening evidence and HPV-site distinctions require care.

Suite links. 14 oncology; 08 infection; 10 rehab.

Evidence anchors: [3,10,27]

5.0.7 Temporomandibular disorders and orofacial pain

Definition and phenotype. Heterogeneous joint, muscle, neuropathic and nociplastic conditions causing jaw pain/dysfunction.

Pathophysiology. Load, injury, inflammation, motor behavior and central pain processes interact.

Determinants and modifiers. Stress, parafunction, trauma, arthritis, sleep and psychosocial factors.

Assessment. Positive history/exam, jaw function and red flags; imaging only for specific joint/structural questions.

Intervention logic. Education, self-management, exercise/splint selectively, pain treatment and minimally invasive/surgical care for defined disease.

Known limits and open questions. Occlusal explanations and irreversible dental treatments are often overstated.

Suite links. 03I MSK; 05 pain/behavior; 06 psychosocial.

Evidence anchors: [4,14,15]

5.0.8 Malocclusion and dentofacial deformity

Definition and phenotype. Tooth/jaw relationships that may affect function, trauma risk, hygiene, airway, appearance or participation.

Pathophysiology. Growth, genetics, habits, tooth loss/eruption and craniofacial development shape alignment.

Determinants and modifiers. Development, congenital conditions, trauma, access and social expectations.

Assessment. Occlusion, growth, function, periodontal risk, imaging and patient-defined goals.

Intervention logic. Observation, orthodontics or orthognathic surgery when benefit justifies burden.

Known limits and open questions. Normal variation can be medicalized and appearance goals may dominate uncertain health benefit.

Suite links. 07 development; 03B airway; 15 ethics.

Evidence anchors: [3,20,36]

5.0.9 Cleft lip/palate and craniofacial differences

Definition and phenotype. Congenital structural differences affecting feeding, speech, hearing, dentition, airway and appearance.

Pathophysiology. Developmental fusion/growth pathways are influenced by genetics and prenatal factors.

Determinants and modifiers. Genetics, selected exposures/nutrition and mostly multifactorial/unknown causes.

Assessment. Multidisciplinary prenatal/postnatal structural, feeding, hearing, speech, dental and developmental assessment.

Intervention logic. Staged surgery, feeding/speech/hearing/dental/psychosocial support across life course.

Known limits and open questions. Optimal timing varies and global access is highly unequal.

Suite links. 03N maternal; 03K hearing; 07 development; 10 rehab.

Evidence anchors: [16,21,22]

5.0.10 Xerostomia and salivary gland disease

Definition and phenotype. Subjective dry mouth and/or objectively reduced or altered saliva from drugs, autoimmune, radiation, obstruction, infection or dehydration.

Pathophysiology. Reduced secretion impairs lubrication, buffering, antimicrobial action and remineralization.

Determinants and modifiers. Polypharmacy, Sjögren disease, radiation, dehydration, diabetes and age.

Assessment. Medication/systemic review, oral effects, standardized flow and imaging/biopsy selectively.

Intervention logic. Treat cause, hydration/stimulation/substitutes, intensive prevention and gland-specific therapy.

Known limits and open questions. Symptoms and flow may disagree; many marketed tests/products have limited evidence.

Suite links. 03H immune; 14 pharmacology; 03O fluid.

Evidence anchors: [3,12]

5.0.11 Tooth wear, erosion, trauma and fracture

Definition and phenotype. Loss or disruption of tooth structure from mechanical, chemical or traumatic forces.

Pathophysiology. Attrition/abrasion, intrinsic/extrinsic acid and impact exceed repair/protection.

Determinants and modifiers. Reflux/eating disorder, diet, occupation, bruxism, sport/violence and anatomy.

Assessment. Pattern, vitality, occlusion, diet/reflux/behavior and imaging.

Intervention logic. Stop cause, protect/preserve pulp and structure, restore when function requires and prevent recurrence.

Known limits and open questions. Attributing wear to bruxism alone is unreliable.

Suite links. 03E reflux; 05 behavior/eating; 09 injury.

Evidence anchors: [4,23,35]

5.0.12 Tooth loss and edentulism

Definition and phenotype. Loss of dentition from caries, periodontal disease, trauma or treatment, affecting chewing, speech and identity.

Pathophysiology. Reduced support and altered loading cause ridge change and functional adaptation.

Determinants and modifiers. Lifetime disease burden, poverty, access, smoking, diabetes and care choices.

Assessment. Remaining tissue, function, nutrition, goals and restorative feasibility.

Intervention logic. Prevention; removable/fixed prostheses or implants with maintenance and realistic risk.

Known limits and open questions. Implants do not remove disease risk and device-centered care can neglect affordability/function.

Suite links. 10 assistive products; 06 social health; 15 equity.

Evidence anchors: [3,17]

5.0.13 Sleep-related oral-airway and breathing interactions

Definition and phenotype. Craniofacial anatomy, tongue/jaw position and neuromuscular control can contribute to snoring and obstructive sleep apnea.

Pathophysiology. Upper-airway collapsibility interacts with obesity, sleep state and ventilatory control.

Determinants and modifiers. Anatomy, obesity, age, alcohol/sedatives and nasal disease.

Assessment. Sleep history and validated sleep testing; dental/airway exam supports but does not diagnose apnea.

Intervention logic. Positive airway pressure, weight/behavior, oral appliances for selected patients and surgery selectively.

Known limits and open questions. Consumer snoring or dental imaging is not a sleep-apnea diagnosis.

Suite links. 03B respiratory; 03D obesity; 03O sleep.

Evidence anchors: [4,18,19]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current oral, dental, and craniofacial health risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.

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Stage	Purpose	Components	Output	Common failure
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

6.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current oral, dental, and craniofacial health risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

6.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

6.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

Evidence anchors: ^[5]

6.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

6.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

6.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

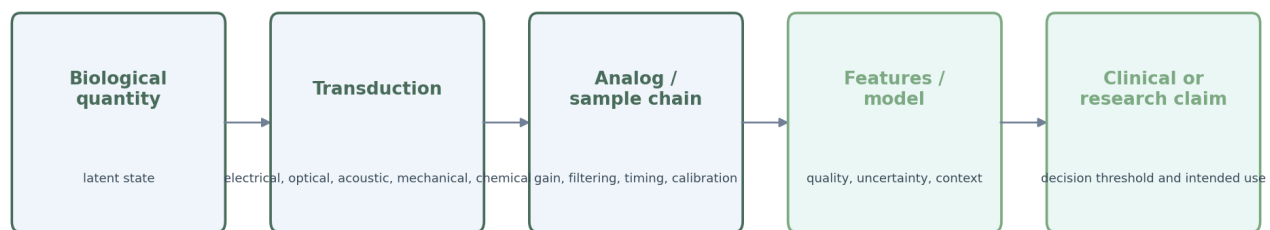
Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

Figure 7.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Clinical oral and dental examination	Noninvasive	Inspection, probing, palpation, percussion, mobility, occlusion and function.	Caries/periodontal/mucosal phenotype and risk.	Direct, low cost and actionable.	Examiner, access, lighting, pain and hidden surfaces.
Intraoral and panoramic radiography	Low-dose ionizing imaging	Projection x-ray attenuation.	Interproximal caries, bone, roots, periapical disease and broad jaw survey.	High value for selected hidden structure.	Radiation, overlap, false positives and inappropriate routine use.
Cone-beam CT	Ionizing 3D imaging	Rotational cone-beam reconstruction of craniofacial hard tissue.	Complex roots, impacted teeth, bone, trauma, implant/surgical planning and selected lesions.	High spatial resolution.	Radiation, artifacts and incidental findings; weak soft-tissue contrast.
Intraoral photography and optical scanning	Noninvasive	Calibrated images or structured-light/confocal 3D surface acquisition.	Lesion documentation, geometry, wear, alignment and restorative design.	Longitudinal, digital and patient-visible.	Moisture, reflectance, stitching, calibration and privacy.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Caries fluorescence/transillumination and electrical methods	Noninvasive	Light scattering/fluorescence or impedance/conductance of tooth tissue.	Adjunct lesion detection or activity proxy.	May detect selected lesions without radiation.	Stain/plaque/moisture, low specificity and device thresholds.
Pulp vitality and sensibility testing	Noninvasive/minimally invasive	Thermally/electrical neural response or optical blood-flow proxy.	Sensibility, probable vitality and comparative tooth response.	Supports endodontic reasoning.	Neural tests are not true perfusion; anxiety, restorations and immature teeth alter response.
Periodontal probing and mobility/force measurement	Minimally invasive contact	Mechanical depth/attachment, bleeding and force/displacement.	Periodontal status, inflammation and support.	Core longitudinal measure.	Probe force/angle, inflammation, landmarks and examiner variability.
Salivary flow and chemistry	Noninvasive sample	Timed secretion, pH/buffer, electrolytes, proteins and microbial/molecular assays.	Gland output and candidate disease/exposure biomarkers.	Low-burden repeated sampling.	Circadian, stimulation, hydration, collection and low specificity.
Microbiology and antimicrobial susceptibility	Minimally invasive sample	Microscopy, culture, PCR or sequencing.	Pathogen/resistance in selected infections.	Useful for severe/refractory disease.	Normal polymicrobial ecology and contamination limit diagnostic interpretation.
Biopsy and oral pathology	Invasive	Histology, immunostains, microbiology and molecular tests.	Cancer, immune, salivary and persistent mucosal diagnosis.	Direct tissue reference.	Sampling/site and procedural harm.
Occlusal force, jaw tracking and muscle EMG	Noninvasive	Pressure/force sensors, motion tracking and surface electrical activity.	Bite force distribution, movement and muscle activation.	Quantifies mechanics.	Sensor alters bite, repeatability, adaptation and weak link to pain.
Swallowing and speech instrumentation	Noninvasive to minimally invasive	Videofluoroscopy, endoscopy, ultrasound, pressure and acoustic analysis.	Bolus safety/efficiency, aspiration and speech articulation.	Direct functional planning.	Radiation/invasiveness, task specificity and observer variation.
Cephalometry and craniofacial 3D analysis	Ionizing/nonionizing imaging	Landmark geometry from radiographs, photography or surface scans.	Growth, asymmetry, airway proxy and surgical/orthodontic planning.	Quantitative structure.	Landmark error, 2D simplification and overclaim about airway/function.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Home toothbrush, aligner and oral-device telemetry	Noninvasive	Motion, pressure, wear-time, temperature or embedded sensor logs.	Adherence and use pattern.	Behavioral feedback and device monitoring.	Gaming, privacy, proprietary algorithms and no proof of health benefit.
Oral optical AI screening	Noninvasive software	Computer vision over intraoral photographs/radiographs.	Caries, lesion, bone or triage score.	Potential access/workflow support.	Dataset shift, labeling, spectrum bias and overdiagnosis.

7.0.1 Clinical oral and dental examination

Measured quantity and transducer. Inspection, probing, palpation, percussion, mobility, occlusion and function.

Typical outputs. Caries/periodontal/mucosal/structural phenotype and risk.

Strengths. Direct, low-cost and actionable.

Limitations and artifacts. Examiner, access, lighting, pain and hidden surfaces.

Reference and validation strategy. Calibration, standardized criteria and longitudinal outcomes.

Clinical/research maturity. Established.

Evidence anchors: [3,4]

7.0.2 Intraoral and panoramic radiography

Measured quantity and transducer. Projection x-ray attenuation.

Typical outputs. Interproximal caries, bone, roots, periapical disease and broad jaw survey.

Strengths. High value for selected hidden structure.

Limitations and artifacts. Radiation, overlap, false positives and inappropriate routine use.

Reference and validation strategy. Dose/geometry QA and indication-based interpretation.

Clinical/research maturity. Established.

Evidence anchors: [4,35]

7.0.3 Cone-beam CT

Measured quantity and transducer. Rotational cone-beam reconstruction of craniofacial hard tissue.

Typical outputs. Complex roots, impacted teeth, bone, trauma, implant/surgical planning and selected lesions.

Strengths. High spatial resolution.

Limitations and artifacts. Radiation, artifacts and incidental findings; weak soft-tissue contrast.

Reference and validation strategy. Protocol/dose optimization and task-specific benefit.

Clinical/research maturity. Established specialist.

Evidence anchors: ^[4,30]

7.0.4 Intraoral photography and optical scanning

Measured quantity and transducer. Calibrated images or structured-light/confocal 3D surface acquisition.

Typical outputs. Lesion documentation, geometry, wear, alignment and restorative design.

Strengths. Longitudinal, digital and patient-visible.

Limitations and artifacts. Moisture, reflectance, stitching, calibration and privacy.

Reference and validation strategy. Reference artifacts/models, repeatability and clinical outcome linkage.

Clinical/research maturity. Established to emerging.

Evidence anchors: ^[28]

7.0.5 Caries fluorescence/transillumination and electrical methods

Measured quantity and transducer. Light scattering/fluorescence or impedance/conductance of tooth tissue.

Typical outputs. Adjunct lesion detection or activity proxy.

Strengths. May detect selected lesions without radiation.

Limitations and artifacts. Stain/plaque/moisture, low specificity and device thresholds.

Reference and validation strategy. Histology/clinical progression and incremental utility over exam/radiographs.

Clinical/research maturity. Adjunct/emerging.

Evidence anchors: [4,30]

7.0.6 Pulp vitality and sensibility testing

Measured quantity and transducer. Thermal/electrical neural response or optical blood-flow proxy.

Typical outputs. Sensibility, probable vitality and comparative tooth response.

Strengths. Supports endodontic reasoning.

Limitations and artifacts. Neural tests are not true perfusion; anxiety, restorations and immature teeth alter response.

Reference and validation strategy. Standard stimulus, contralateral controls and longitudinal tissue outcome.

Clinical/research maturity. Established; perfusion methods emerging.

Evidence anchors: [4,28]

7.0.7 Periodontal probing and mobility/force measurement

Measured quantity and transducer. Mechanical depth/attachment, bleeding and force/displacement.

Typical outputs. Periodontal status, inflammation and support.

Strengths. Core longitudinal measure.

Limitations and artifacts. Probe force/angle, inflammation, landmarks and examiner variability.

Reference and validation strategy. Calibration, reproducibility and radiographic/clinical outcome.

Clinical/research maturity. Established.

Evidence anchors: [4]

7.0.8 Salivary flow and chemistry

Measured quantity and transducer. Timed secretion, pH/buffer, electrolytes, proteins and microbial/molecular assays.

Typical outputs. Gland output and candidate disease/exposure biomarkers.

Strengths. Low-burden repeated sampling.

Limitations and artifacts. Circadian, stimulation, hydration, collection and low specificity.

Reference and validation strategy. Standard preanalytics, assay validation and utility studies.

Clinical/research maturity. Established for flow; biomarkers mostly research.

Evidence anchors: ^[3]

7.0.9 Microbiology and antimicrobial susceptibility

Measured quantity and transducer. Microscopy, culture, PCR or sequencing.

Typical outputs. Pathogen/resistance in selected infections.

Strengths. Useful for severe/refractory disease.

Limitations and artifacts. Normal polymicrobial ecology and contamination limit diagnostic interpretation.

Reference and validation strategy. Syndrome-linked specimen and stewardship outcomes.

Clinical/research maturity. Established selectively.

Evidence anchors: ^[8,9]

7.0.10 Biopsy and oral pathology

Measured quantity and transducer. Histology, immunostains, microbiology and molecular tests.

Typical outputs. Cancer, immune, salivary and persistent mucosal diagnosis.

Strengths. Direct tissue reference.

Limitations and artifacts. Sampling/site and procedural harm.

Reference and validation strategy. Adequate lesion edge/depth and clinicopathologic correlation.

Clinical/research maturity. Established.

Evidence anchors: ^[5,10]

7.0.11 Occlusal force, jaw tracking and muscle EMG

Measured quantity and transducer. Pressure/force sensors, motion tracking and surface electrical activity.

Typical outputs. Bite force distribution, movement and muscle activation.

Strengths. Quantifies mechanics.

Limitations and artifacts. Sensor alters bite, repeatability, adaptation and weak link to pain.

Reference and validation strategy. Bench calibration, standardized tasks and outcome-based validity.

Clinical/research maturity. Research/selected clinical.

Evidence anchors: ^[15,31]

7.0.12 Swallowing and speech instrumentation

Measured quantity and transducer. Videofluoroscopy, endoscopy, ultrasound, pressure and acoustic analysis.

Typical outputs. Bolus safety/efficiency, aspiration and speech articulation.

Strengths. Direct functional planning.

Limitations and artifacts. Radiation/invasiveness, task specificity and observer variation.

Reference and validation strategy. Standard protocols and clinical outcome correlation.

Clinical/research maturity. Established specialist.

Evidence anchors: ^[16]

7.0.13 Cephalometry and craniofacial 3D analysis

Measured quantity and transducer. Landmark geometry from radiographs, photography or surface scans.

Typical outputs. Growth, asymmetry, airway proxy and surgical/orthodontic planning.

Strengths. Quantitative structure.

Limitations and artifacts. Landmark error, 2D simplification and overclaim about airway/function.

Reference and validation strategy. Reproducibility and functional outcome studies.

Clinical/research maturity. Established planning; limited diagnostic claims.

Evidence anchors: ^[4,33]

7.0.14 Home toothbrush, aligner and oral-device telemetry

Measured quantity and transducer. Motion, pressure, wear-time, temperature or embedded sensor logs.

Typical outputs. Adherence and use pattern.

Strengths. Behavioral feedback and device monitoring.

Limitations and artifacts. Gaming, privacy, proprietary algorithms and no proof of health benefit.

Reference and validation strategy. Bench/usability, objective reference and randomized utility.

Clinical/research maturity. Emerging.

Evidence anchors: [28,32]

7.0.15 Oral optical AI screening

Measured quantity and transducer. Computer vision over intraoral photographs/radiographs.

Typical outputs. Caries, lesion, bone or triage score.

Strengths. Potential access/workflow support.

Limitations and artifacts. Dataset shift, labeling, spectrum bias and overdiagnosis.

Reference and validation strategy. Prospective external and subgroup validation plus utility.

Clinical/research maturity. Emerging.

Evidence anchors: [29,33,34]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise oral, dental, and craniofacial health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

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Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

8.0.1 Intended use and measurand

Design requirements. Define the precise oral, dental, and craniofacial health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: ^[28]

8.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: ^[31]

8.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

8.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

Evidence anchors: ^[30]

8.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[30]

8.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: [33,34]

8.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: [29,33]

8.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: [31,32]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Fluoride, sealants and risk-based prevention	Prevention	Increase remineralization and reduce lesion initiation/progression.	Strong population and individual evidence for caries prevention.	Fluorosis only with excess developmental ingestion; access and adherence vary.	Lesion activity, exposure, new disease and equity.
Diet and biofilm self-care support	Prevention	Reduce frequent fermentable substrate and disrupt harmful plaque while preserving feasible nutrition.	Foundational for caries/periodontal prevention.	Blame, eating-disorder risk and structural food/access constraints.	Behavior, disease trajectory and patient burden.
Professional periodontal and restorative care	Treatment	Remove diseased tissue/biofilm and restore form/function with minimally invasive principles.	Effective when matched to lesion activity, periodontal stage and maintenance.	Procedural harm, overtreatment, material failure and cost.	Disease control, tooth survival, symptoms, function and maintenance.
Endodontic therapy and extraction	Definitive/supportive control	Preserve or remove teeth with irreversible pulp/apical disease.	High success in selected teeth with appropriate restoration and follow-up.	Pain, procedural injury, fracture, retreatment and tooth loss consequences.	Healing, symptoms, function and restoration integrity.
Antimicrobial stewardship	Treatment/syndrome	Preserve antibiotics for spreading/systemic infection and use source control/local care when sufficient.	Reduces avoidable exposure without compromising appropriate care.	Resistance, allergy and treatment delay if severity missed.	Indication, duration, clinical response and prescribing patterns.
Orthodontic and craniofacial treatment	Developmental	Guide structural alignment/growth or surgically correct deformity to meet functional and person-valued goals.	Effective for defined malocclusion/deformity; health benefit varies by indication.	Root resorption, caries, periodontal effects, surgery and long burden.	Function, hygiene, stability, QOL and adverse effects.

Continued on next page

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Prosthodontics and implants	Assistive/structure	Replace missing teeth with removable, fixed or implant-supported systems.	Improves chewing/appearance and selected quality outcomes.	Peri-implant disease, failure, maintenance, surgery and affordability.	Use, comfort, diet/function, tissue health and maintenance.
Oral cancer prevention and treatment	Prevention/orology	Reduce carcinogenic exposures and treat lesions with surgery/radiation/systemic therapy.	Exposure control and stage-appropriate multidisciplinary care improve outcomes.	Major speech/swallowing effects and access inequity.	Recurrence/survival, appearance speech/swallow, pain and participation.
TMD/orofacial pain rehabilitation	Whole-person	Education, jaw exercise, behavioral/sleep and selected dental/medical interventions.	Conservative reversible care benefits many; invasive occlusal approaches lack broad support.	Persistent pain, stigma and overtreatment.	Jaw function, pain interference, sleep and participation.
Salivary and mucosal supportive care	Supportive	Stimulate/substitute saliva, protect mucosa, control infection and maintain nutrition/oral hygiene.	Important in radiation, autoimmune and medication-related states.	Product burden and inadequate disease control.	Flow/symptoms, caries/infection, eating/speech and adherence.
Population oral-health systems	Public health/delivery	Integrate prevention, primary care, school/community programs and affordable essential dental services.	Addresses highly preventable disease and access gaps.	Workforce, financing and fragmented medical-dental records.	Coverage, untreated disease, pain, school/work loss, equity and catastrophic spending.

9.0.1 Fluoride, sealants and risk-based prevention

Target and mechanism. Increase remineralization and reduce lesion initiation/progression.

Evidence and expected outcome. Strong population and individual evidence for caries prevention.

Risks, burden, and equity. Fluorosis only with excess developmental ingestion; access and adherence vary.

Response monitoring. Lesion activity, exposure, new disease and equity.

Evidence anchors: [3,4]

9.0.2 Diet and biofilm self-care support

Target and mechanism. Reduce frequent fermentable substrate and disrupt harmful plaque while preserving feasible nutrition.

Evidence and expected outcome. Foundational for caries/periodontal prevention.

Risks, burden, and equity. Blame, eating-disorder risk and structural food/access constraints.

Response monitoring. Behavior, disease trajectory and patient burden.

Evidence anchors: [3,24]

9.0.3 Professional periodontal and restorative care

Target and mechanism. Remove diseased tissue/biofilm and restore form/function with minimally invasive principles.

Evidence and expected outcome. Effective when matched to lesion activity, periodontal stage and maintenance.

Risks, burden, and equity. Procedural harm, overtreatment, material failure and cost.

Response monitoring. Disease control, tooth survival, symptoms, function and maintenance.

Evidence anchors: [4]

9.0.4 Endodontic therapy and extraction

Target and mechanism. Preserve or remove teeth with irreversible pulp/apical disease.

Evidence and expected outcome. High success in selected teeth with appropriate restoration and follow-up.

Risks, burden, and equity. Pain, procedural injury, fracture, retreatment and tooth loss consequences.

Response monitoring. Healing, symptoms, function and restoration integrity.

Evidence anchors: [4,35]

9.0.5 Antimicrobial stewardship

Target and mechanism. Reserve antibiotics for spreading/systemic infection and use source control/local care when sufficient.

Evidence and expected outcome. Reduces avoidable exposure without compromising appropriate care.

Risks, burden, and equity. Resistance, allergy and treatment delay if severity missed.

Response monitoring. Indication, duration, clinical response and prescribing patterns.

Evidence anchors: [8,9]

9.0.6 Orthodontic and craniofacial treatment

Target and mechanism. Guide alignment/growth or surgically correct deformity to meet functional and person-valued goals.

Evidence and expected outcome. Effective for defined malocclusion/deformity; health benefit varies by indication.

Risks, burden, and equity. Root resorption, caries, periodontal effects, surgery and long burden.

Response monitoring. Function, hygiene, stability, QOL and adverse effects.

Evidence anchors: [4,36]

9.0.7 Prosthodontics and implants

Target and mechanism. Replace missing teeth with removable, fixed or implant-supported systems.

Evidence and expected outcome. Improves chewing/appearance and selected quality outcomes.

Risks, burden, and equity. Peri-implant disease, failure, maintenance, surgery and affordability.

Response monitoring. Use, comfort, diet/function, tissue health and maintenance.

Evidence anchors: [3,17]

9.0.8 Oral cancer prevention and treatment

Target and mechanism. Reduce carcinogenic exposures and treat lesions with surgery/radiation/systemic therapy.

Evidence and expected outcome. Exposure control and stage-appropriate multidisciplinary care improve outcomes.

Risks, burden, and equity. Major speech/swallowing/appearance effects and access inequity.

Response monitoring. Recurrence/survival, nutrition, speech/swallow, pain and participation.

Evidence anchors: [3,10,27]

9.0.9 TMD/orofacial pain rehabilitation

Target and mechanism. Education, jaw exercise, behavioral/sleep and selected dental/medical interventions.

Evidence and expected outcome. Conservative reversible care benefits many; invasive occlusal approaches lack broad support.

Risks, burden, and equity. Persistent pain, stigma and overtreatment.

Response monitoring. Jaw function, pain interference, sleep and participation.

Evidence anchors: [14,16]

9.0.10 Salivary and mucosal supportive care

Target and mechanism. Stimulate/substitute saliva, protect mucosa, control infection and maintain nutrition/oral hygiene.

Evidence and expected outcome. Important in radiation, autoimmune and medication-related states.

Risks, burden, and equity. Product burden and inadequate disease control.

Response monitoring. Flow/symptoms, caries/infection, eating/speech and adherence.

Evidence anchors: [3,27]

9.0.11 Population oral-health systems

Target and mechanism. Integrate prevention, primary care, school/community programs and affordable essential dental services.

Evidence and expected outcome. Addresses highly preventable disease and access gaps.

Risks, burden, and equity. Workforce, financing and fragmented medical-dental records.

Response monitoring. Coverage, untreated disease, pain, school/work loss, equity and catastrophic spending.

Evidence anchors: [3,25,26]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Does treating periodontal disease prevent major systemic events?	Periodontal and systemic disease share inflammatory and behavioral determinants; periodontal treatment improves oral outcomes and some intermediate markers.	Magnitude of causal reduction in cardiovascular, pregnancy or metabolic endpoints.	Confounding, heterogeneous interventions and surrogate endpoints.	Large pragmatic trials or strong triangulation with patient-important systemic outcomes.
Can saliva support broad disease screening?	Saliva contains host, microbial and exposure molecules.	Reproducible incremental utility for systemic screening beyond simpler pathways.	Collection and oral disease strongly alter signals; low-prevalence false positives.	Locked multisite assays and randomized diagnostic-utility studies.
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within oral, dental, and craniofacial health.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

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Question	What is established	What remains uncertain	Why unresolved	Decisive next step
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

10.0.1 Does treating periodontal disease prevent major systemic events?

Established knowledge. Periodontal and systemic disease share inflammatory and behavioral determinants; periodontal treatment improves oral outcomes and some intermediate markers.

Unresolved issue. Magnitude of causal reduction in cardiovascular, pregnancy or metabolic endpoints.

Why current evidence is insufficient. Confounding, heterogeneous interventions and surrogate endpoints.

Research that would change confidence. Large pragmatic trials or strong triangulation with patient-important systemic outcomes.

Evidence anchors: ^[2]

10.0.2 Can saliva support broad disease screening?

Established knowledge. Saliva contains host, microbial and exposure molecules.

Unresolved issue. Reproducible incremental utility for systemic screening beyond simpler pathways.

Why current evidence is insufficient. Collection and oral disease strongly alter signals; low-prevalence false positives.

Research that would change confidence. Locked multisite assays and randomized diagnostic-utility studies.

Evidence anchors: ^[33]

10.0.3 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within oral, dental, and craniofacial health.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

10.0.4 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: ^[33]

10.0.5 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[10]

10.0.6 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[28,34]

10.0.7 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Integrated oral-systemic cohort	Which oral trajectories predict function and shared systemic risk after confounding control?	Longitudinal dental, medical, diet, smoking, social and microbiome data with standardized periodontal/caries measures.	Causal estimates that identify effective integrated interventions.	Association-only claims used to market testing.
0. Low-cost optical caries/lesion platform	Can calibrated imaging improve detection without overdiagnosis?	Bench optical standards, diverse prospective clinical cohort and pathology/progression reference where appropriate.	Improved patient-important decisions with bounded false positives.	Higher sensitivity increases unnecessary procedures.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for oral, dental, and craniofacial health?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.

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Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

11.0.1 0. Integrated oral-systemic cohort

Question. Which oral trajectories predict function and shared systemic risk after confounding control?

Design. Longitudinal dental, medical, diet, smoking, social and microbiome data with standardized periodontal/caries measures.

Success criterion. Causal estimates that identify effective integrated interventions.

Failure or stop criterion. Association-only claims used to market testing.

11.0.2 0. Low-cost optical caries/lesion platform

Question. Can calibrated imaging improve detection without overdiagnosis?

Design. Bench optical standards, diverse prospective clinical cohort and pathology/progression reference where appropriate.

Success criterion. Improved patient-important decisions with bounded false positives.

Failure or stop criterion. Higher sensitivity increases unnecessary procedures.

Evidence anchors: ^[28]

11.0.3 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for oral, dental, and craniofacial health?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

11.0.4 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

11.0.5 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: ^[33]

11.0.6 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

11.0.7 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

11.0.8 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03E	Gastrointestinal Health	Nutrition, reflux, swallowing and microbiome connect oral and digestive systems.	tooth wear, dysphagia or nutrition is central.
03D	Endocrine and Metabolic Health	Diabetes, obesity and endocrine disorders alter periodontal disease, infection and saliva.	healing or disease activity is unexpectedly poor.
03B	Respiratory Health	Airway anatomy, aspiration and sleep-disordered breathing bridge domains.	oral/dental devices are proposed for sleep or airway claims.
03H	Immune and Lymphatic Health	Mucosal immunity, autoimmune salivary disease and infection are shared.	ulcers, xerostomia or periodontal inflammation is systemic.
10	Function, Disability and Rehabilitation	Feeding, speech, swallowing, prostheses and craniofacial rehabilitation determine participation.	structural treatment must translate to daily function.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Caries	Biofilm-mediated, sugar-driven, multifactorial, dynamic disease causing net mineral loss in dental hard tissues.
Periodontium	Gingiva, periodontal ligament, cementum and alveolar bone supporting teeth.
Xerostomia	Subjective dry mouth; may occur with or without objectively low salivary flow.
Occlusion	Contact and functional relationship between upper and lower teeth/jaws.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.